AIN SHAMS UNIVERSITY

FACULTY OF ENGINEERING

MECHATRONICS ENGINEERING DEPARTMENT

Spring 2026

MCT344 / MCT342 Industrial Robotics

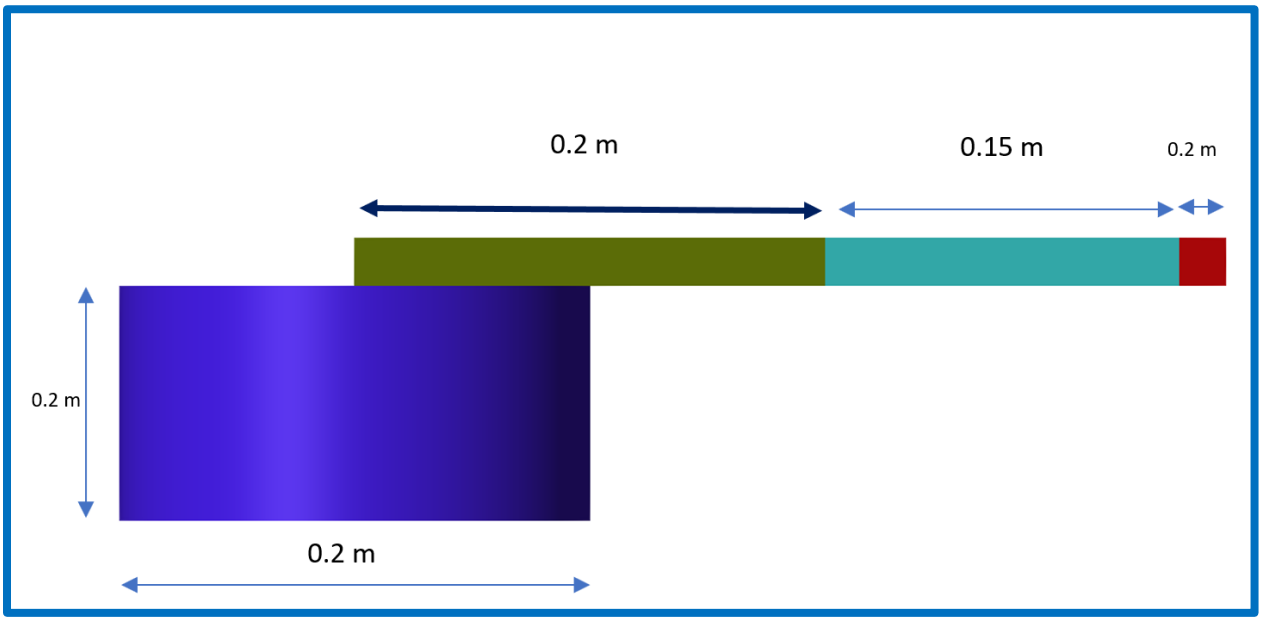
Lab Sheet 1

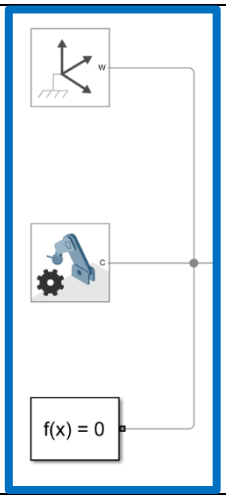
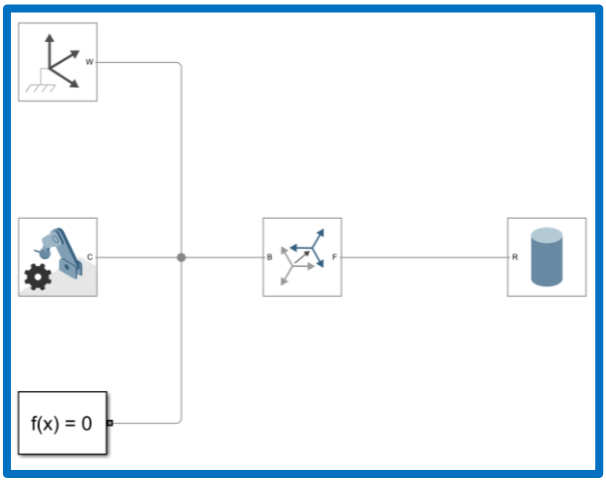
MATLAB & Simulink

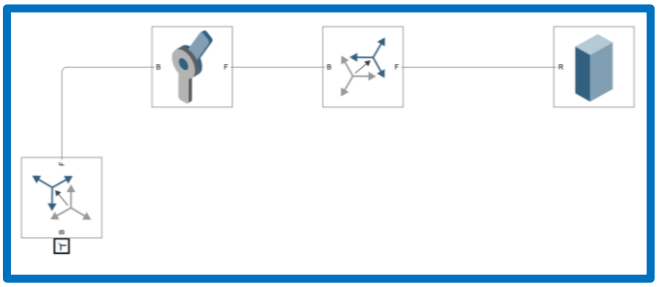
Lab1: Robot Creation and Forward Kinematics

In this lab, you are required to create the following RR Robot with these dimensions using Simscape tool box, and apply forward kinematics to it.

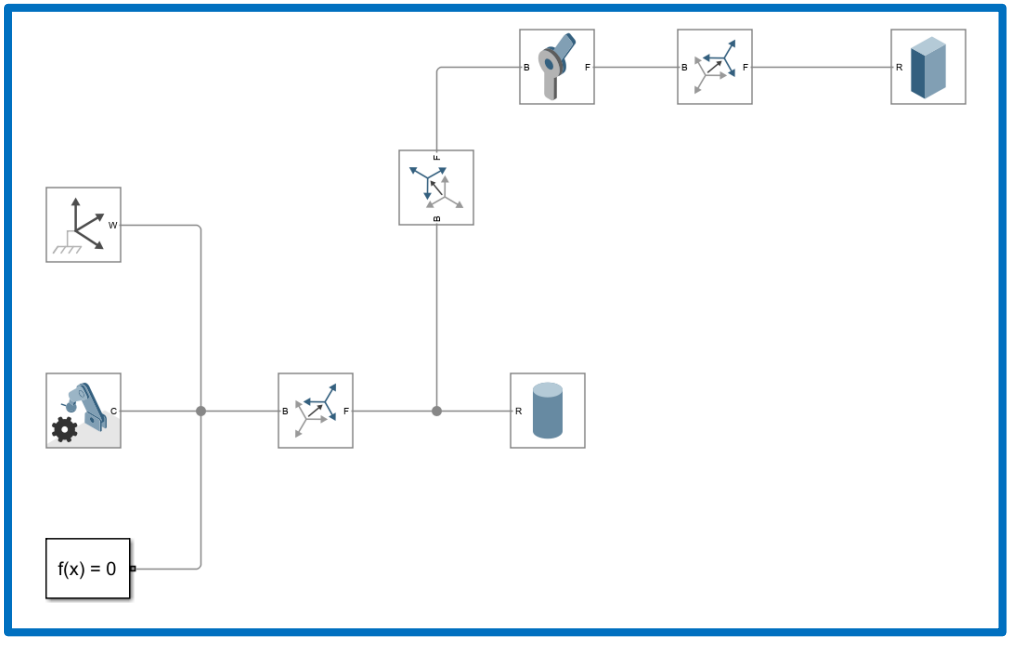
Part 1: Using Simscape Toolbox to Create Robot



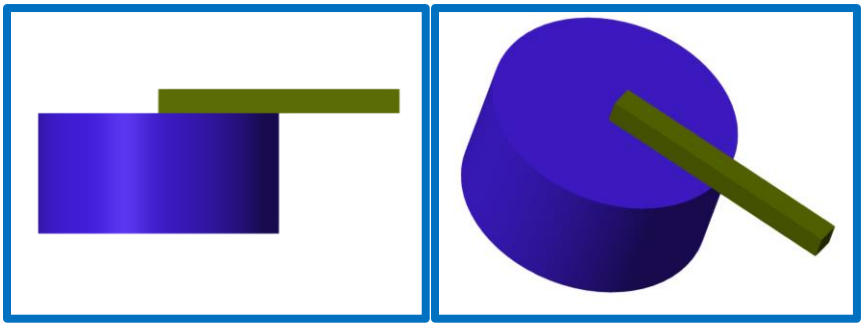
Step		Action	Screenshot
1		Simscape > Utilities > **Solver Configuration** Simscape > Multibody > Frames and Transforms > **World Frame** Simscape > Multibody > Utilities > **Mechanism Configuration**	
2	A	Simscape > Multibody > Body Element > **Cylindrical Solid** **Cylindrical Solid Block:** Modify length, radius and color **Note:** By default, the frame of any rigid body we create is at its center	
	B	Simscape > Multibody > Frames and Transforms > **Rigid Transform** **Rigid Transform Block:** B === World frame F === Cylindrical Solid Block Translation > Method = Cartesian Modify offset to match half of the height of the cylinder, so that the base of the cylinder is setting on the world frame	

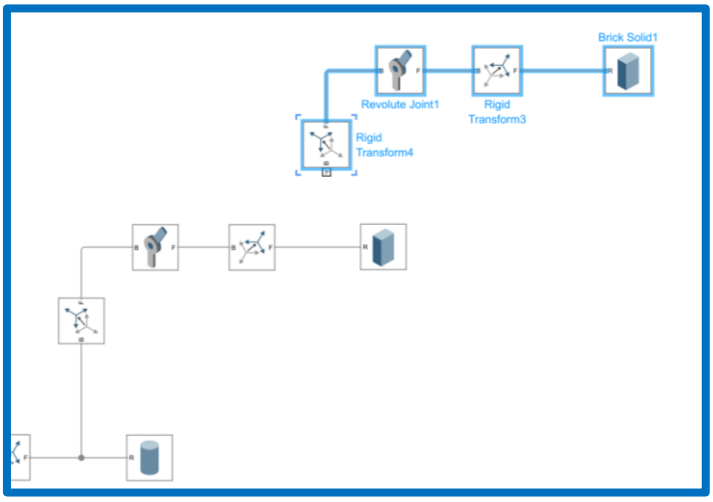
Step		Action	Screenshot
3	A	Simscape > Multibody > Body Element > **Brick Solid** **Brick Block:** Modify length, width, height and color	
	B	Simscape > Multibody > Joints > **Revolute Joint** **Note:** Revolute Joints by default has Z-axis pointing upwards and it's the axis of rotation.	
	C	**Rigid Transform Block:** **(Between Brick and Revolute Joint)** Modify offset to match half of the length of the link, so that the joint is at the end of the link.	
	D	**Rigid Transform Block:** **(Between Revolute Joint and the base (Cylinder))** Modify offset to match half of the height of the base + half the height of the link (Brick), so that the link is above the base.	

Currently, you should have something like this:

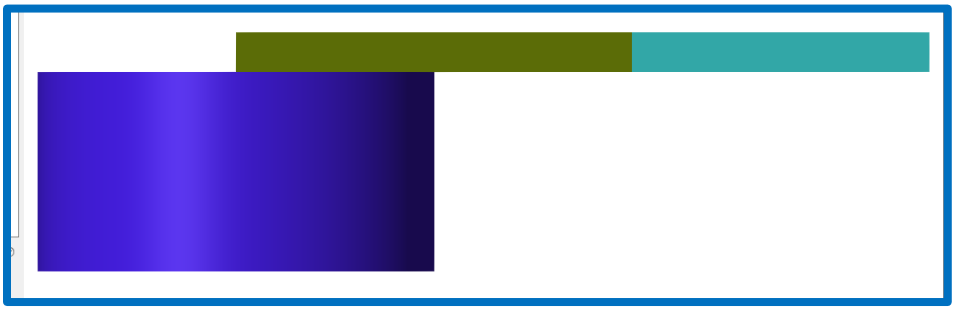


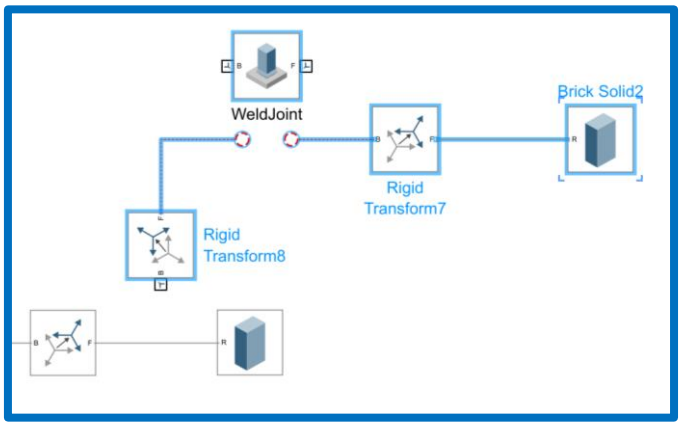
When you run, you should have something like this:



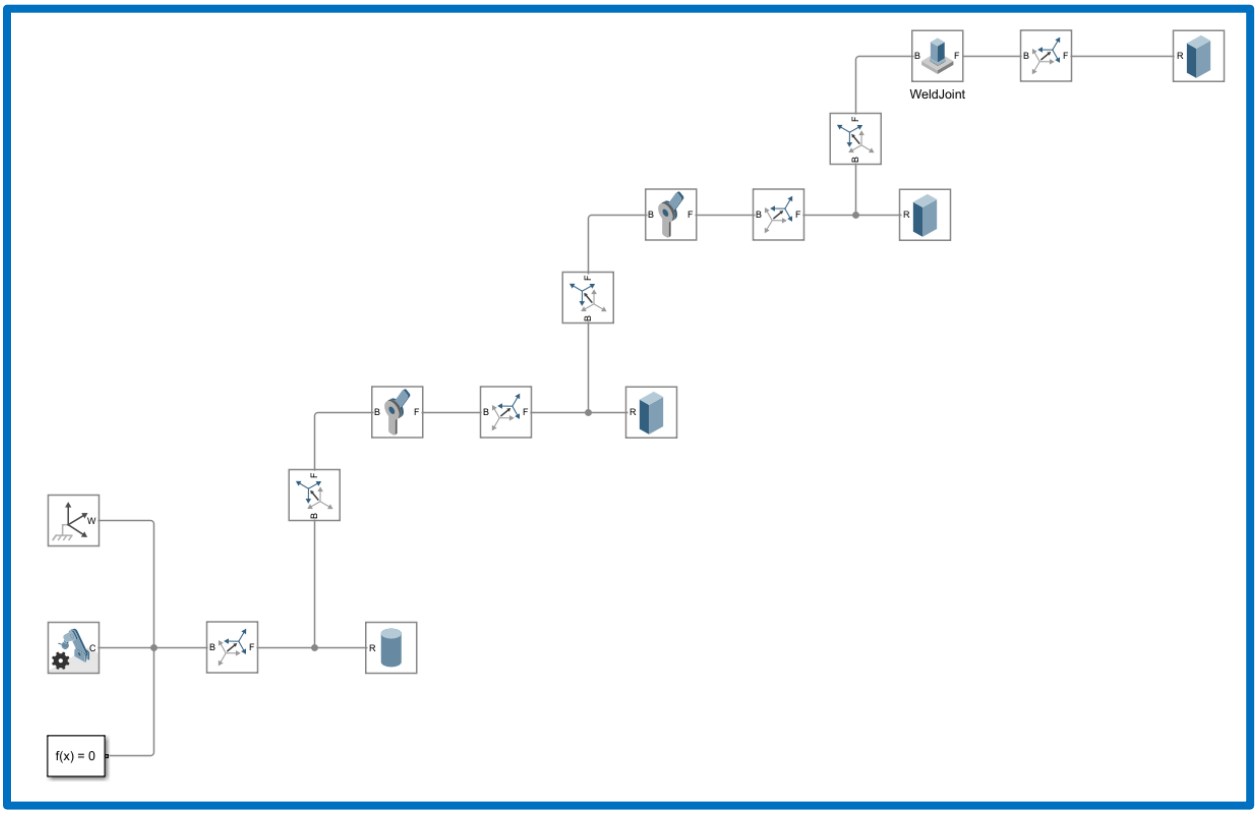
Step		Action	Screenshot
4	A	Copy These 4 Blocks	
	B	**Brick Block:** Modify length and color only	
	C	**Rigid Transform Block: (Between Revolute Joint and link 1 (Brick))** Modify offset to match half of the length of the link.	
	D	**Rigid Transform Block: (Between Revolute Joint and Brick 2)** Modify offset to match half of the length of the link 2	

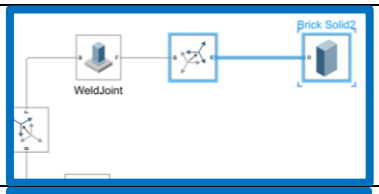
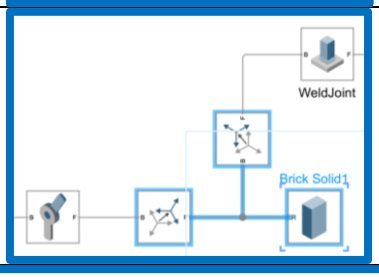
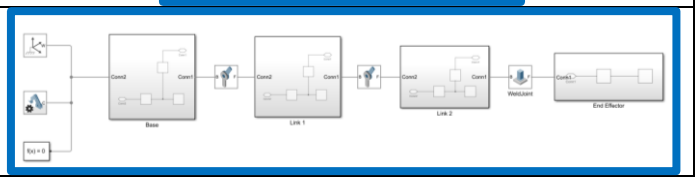
Currently, when you run, you should have something like this:



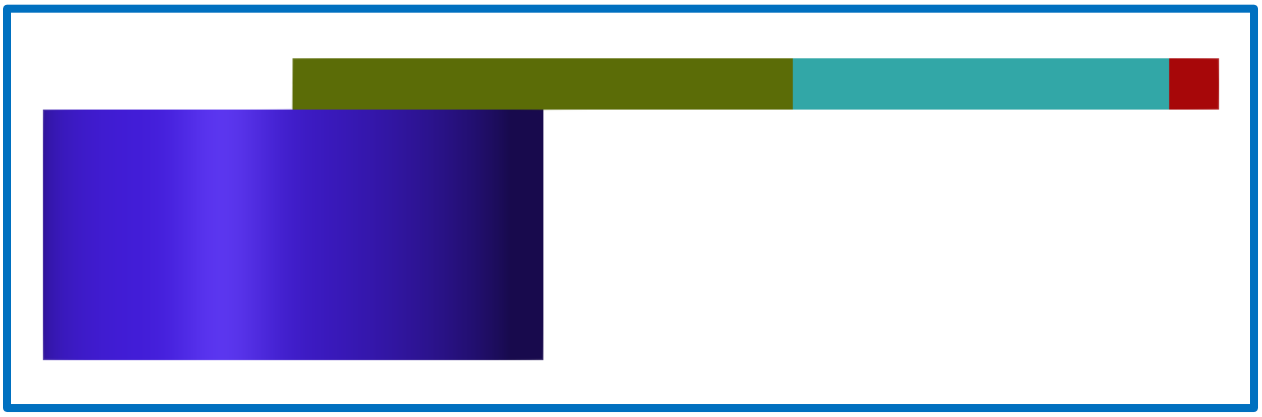
Step		Action	Screenshot
5	A	Copy These 4 Blocks Simscape > Multibody > Joints > **Weld Joint** Replace **Revolute** Joint with **Weld** Joint	
	B	**Brick Block:** Modify length and color only	
	C	**Rigid Transform Block: (Between Revolute Joint and link 2 (Brick))** Modify offset to match half of the length of the link.	
	D	**Rigid Transform Block: (Between Revolute Joint and Brick 3)** Modify offset to match half of the length of the brick 3	

In the End, you should have something like this:



Step		Action	Screenshot
6	A	Group Each Link in a Subsystem for a more clear Visual using the 3 dots when you select blocks	 
	B	We want to reach this simplified version:	

When you run, you should have something like this:

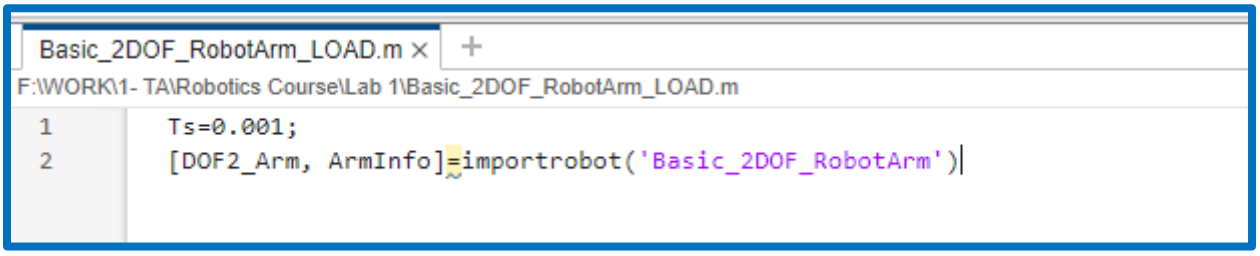


Part 2: Using Robotic Systems Toolbox to Apply Forward Kinematics

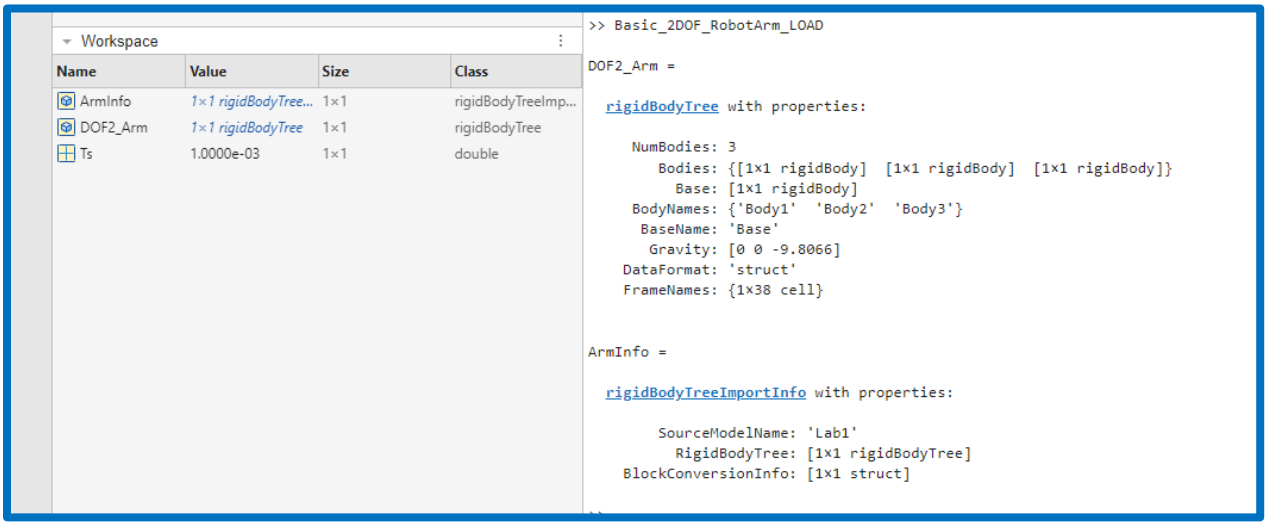
First, we need to create a Rigid Body Tree representation for our robot.

To do this we will use MATLAB:

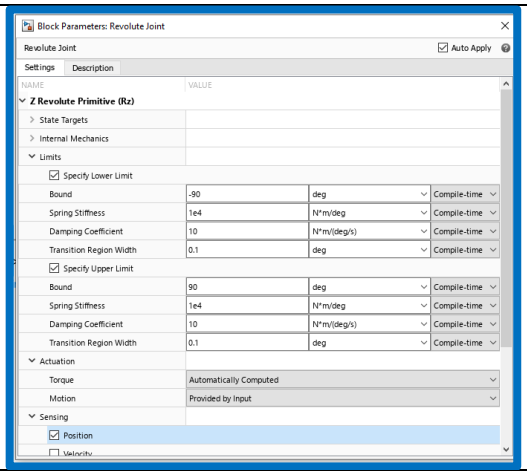
Creating a MATLAB (.m) file
MATLAB Home Tab > New > Script
`Ts=0.001;` `[DOF2_Arm, ArmInfo]=importrobot(*** YOUR Simulink model file NAME ***)`

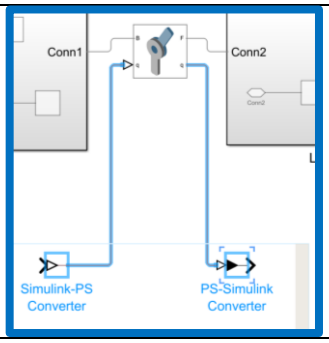
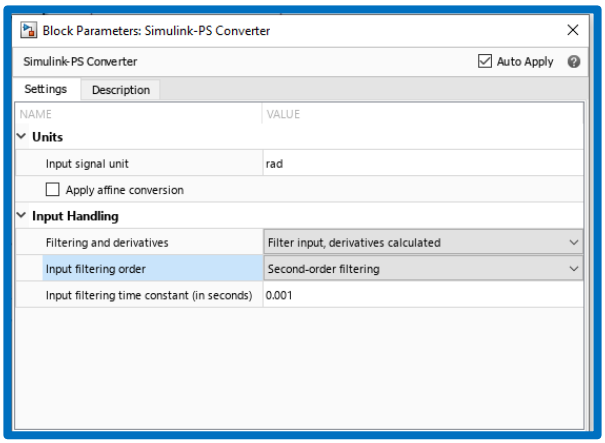


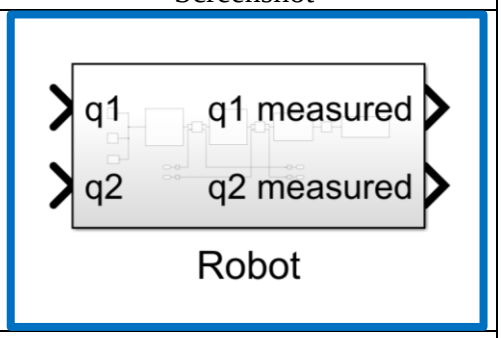
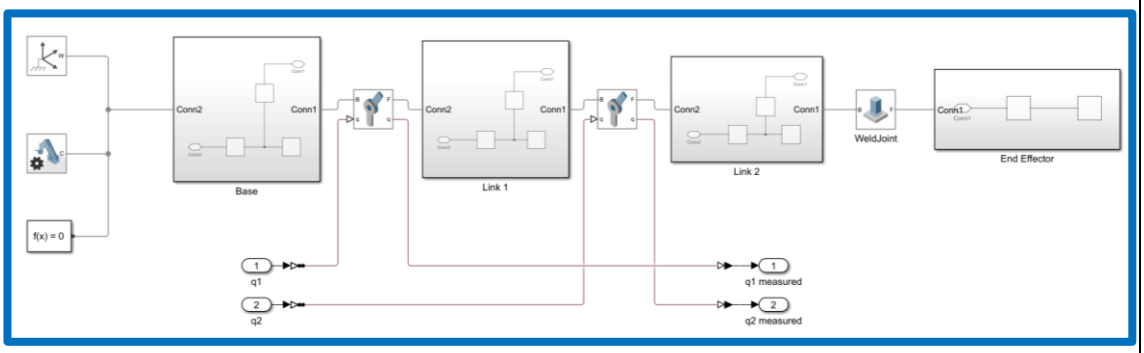
This should appear:

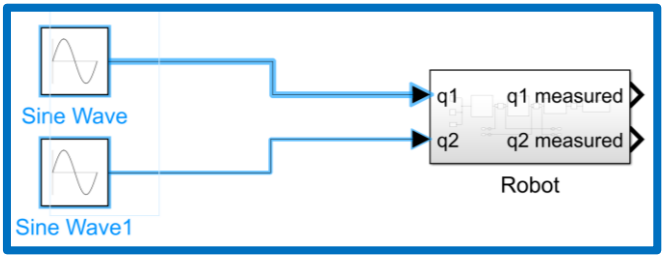


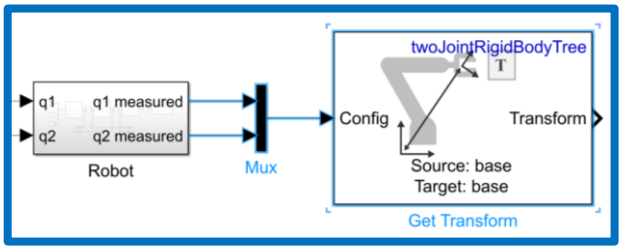
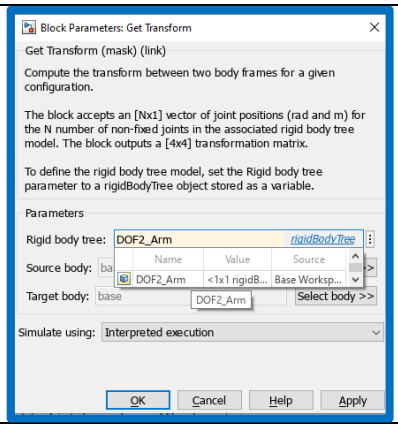
Second, we prepare the joints to be able to take input from us:

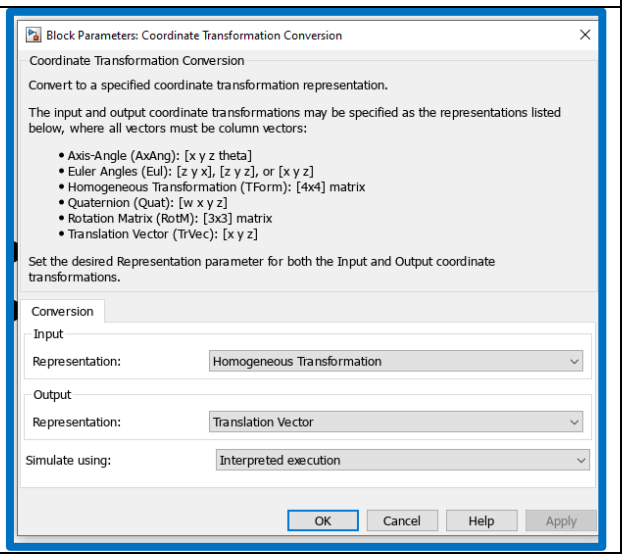
Step		Action	Screenshot
2	A	Open the revolute Joint Block	
	B	**Revolute Joint Block:** **Limits Tab:** Specify Lower Limit > -90 Specify Upper Limit > 90	
	C	**Revolute Joint Block:** **Actuation Tab:** Motion >> Provided by input Torque >> Automatically Computed	
	D	**Revolute Joint Block:** **Sensing Tab:** Position >> Check	
	E	Then, Repeat the process for the other Revolute Joint.	

Step		Action	Screenshot
3	A	Simscape Toolbox >> Utilities >> PS-Simulink Converter Simscape Toolbox >> Utilities >> Simulink-PS Converter	
	B	**Simulink-PS Converter Block:** **Units Tab:** Input Signal Unit >> Rad	
	C	**Simulink-PS Converter Block:** **Input-Handling Tab:** Filtering and Derivatives >> Filter input, Derivative calculated Input Filtering Order >> Second Order Filtering Input Filtering Time Constant >> 0.001	
	D	Then, Copy the 2 converters, for the other Revolute Joint	

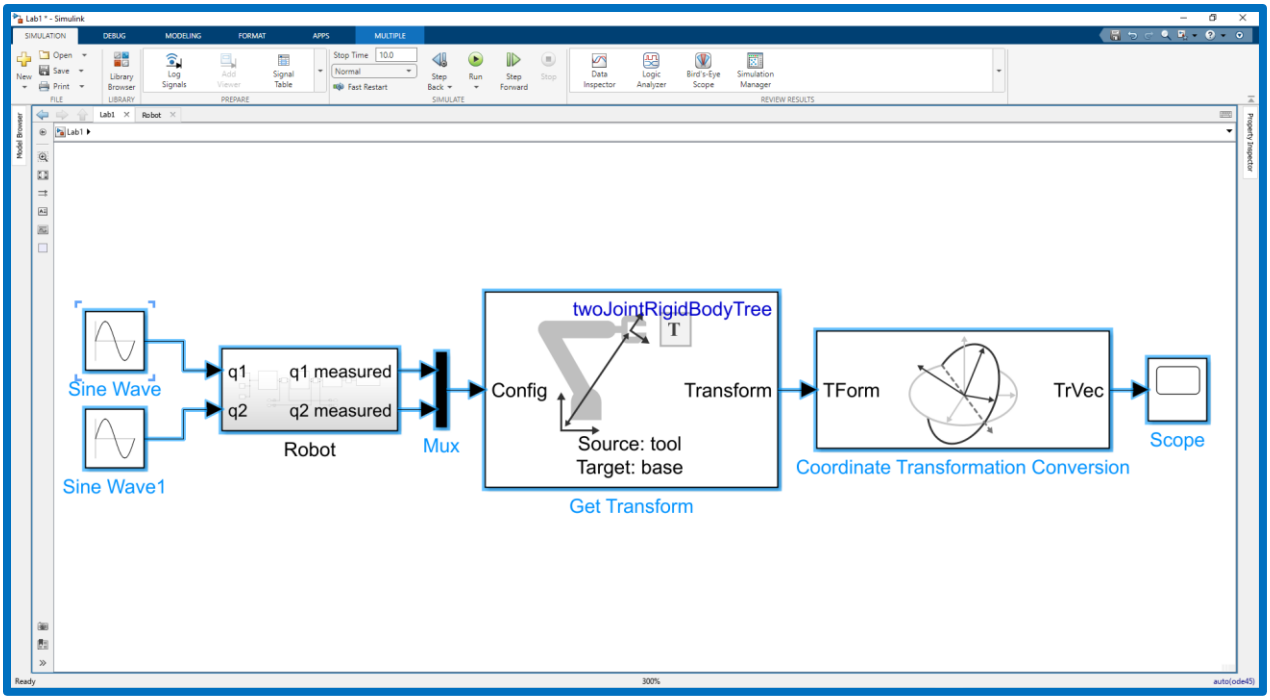
Step		Action	Screenshot
4	A	Select the Whole model now and make it one subsystem called “Robot”	
	B	**Rename the Inputs and Outputs (the converters Ports) as Follows:** Rev Joint 1 Input: q1 Rev Joint 2 Input: q2 Rev Joint 1 Output: q1 measured Rev Joint 2 Output: q2 measured	
	C		

Step		Action	Screenshot
5	A	Insert Sine wave	
	B	**Sine wave Block 1:** **Amplitude** >> (Must be less than 90 degrees, but in radians, Less than Pi /2) >> for example: 1.3 **Frequency** >> for example 1	
	C	**Sine wave Block 2:** **Amplitude** >> (Must be less than 90 degrees, but in radians, Less than Pi /2) >> for example: 0.6 **Frequency** >> for example 0.7	
	D	Connect both to q1 & q2	

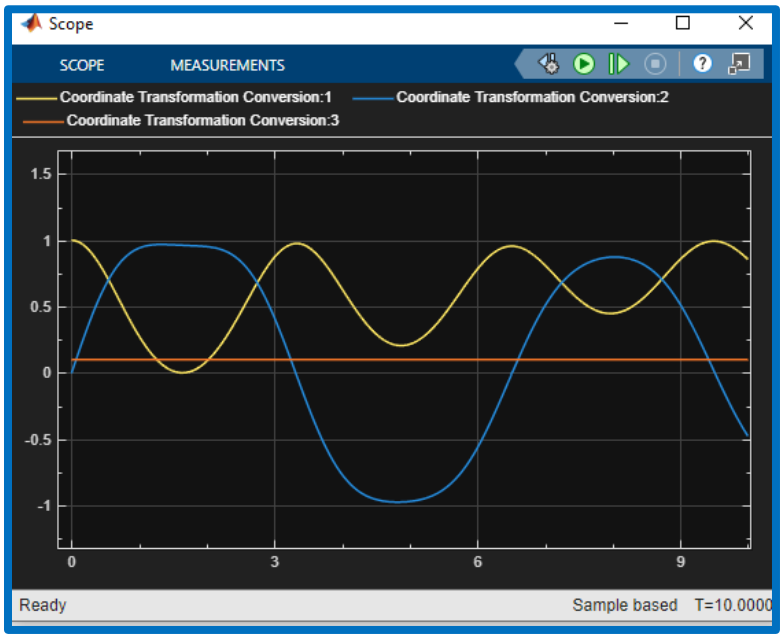
Step		Action	Screenshot
6	A	Robotic Systems Toolbox > Manipulator Algorithms > Get Transform **Note: Get Transform Needs:** 1- Configuration 2- Rigid Body Tree	
	B	**1- Configuration:** Insert >> Mux Combine q1m and q2m using the Mux, then provide it to Get Transform Block	
	C	**2- Rigid Body Tree** **Get Transform Block:** Rigid Body Tree >> DOF2_Arm ***(the one we created using MATLAB .m file)***	
	D	**Get Transform Block:** Source Body >> Body3 ***(the LAST body, The End Effector)***	
	E	**Get Transform Block:** Target Body >> Base ***(the FIRST body, The Base)***	

Step		Action	Screenshot
7	A	Robotic Systems Toolbox > Utilities > Coordinate Transformation Conversion	
	B	**Coordinate Transformation Conversion Block:** Input Representation >> Homogenous Transformation Output Representation >> Translation Vector	
	C	Take the output of this block to a scope	

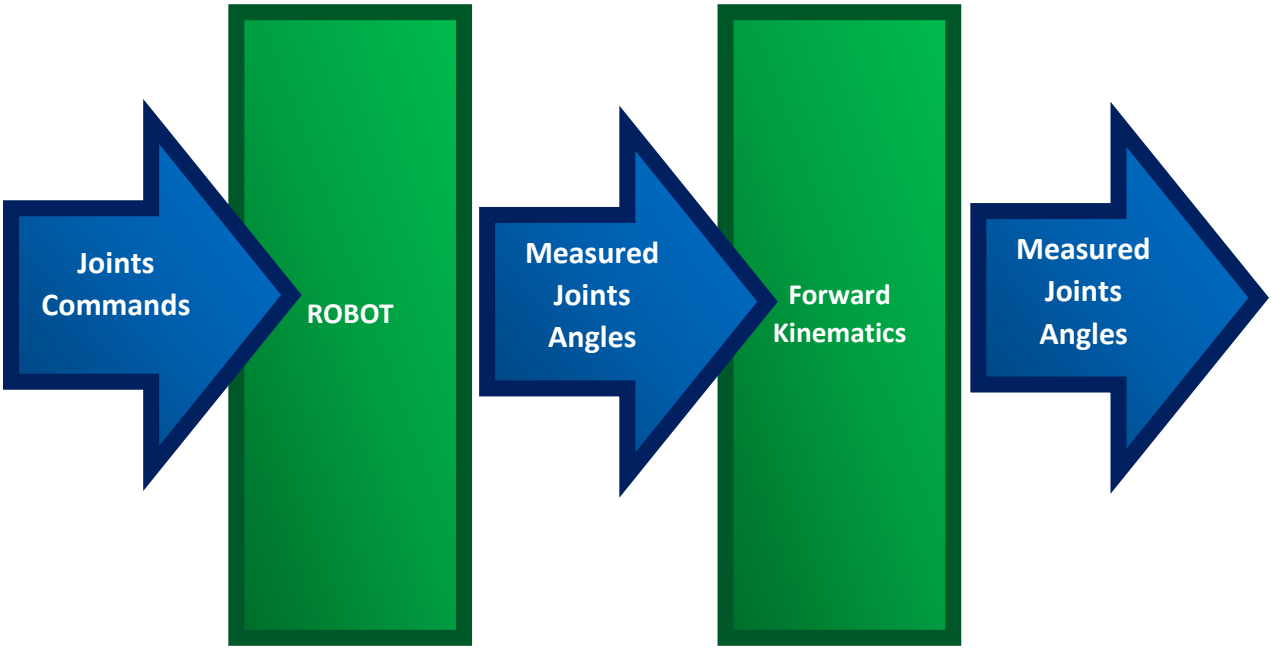
Now Your Model should look like this:



And the Scope should display something like this:



Summary


Joints Commands
ROBOT
Measured Joints Angles
Forward Kinematics
Measured Joints Angles